

Euclid's Elements — Book I

Proposition 33

Formal Reconstruction — Technical Documentation

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Joining equal and parallel segments

1 Introduction

Euclid's Proposition I.33 states that the straight lines joining the corresponding ends of equal and parallel straight lines in the same directions are themselves equal and parallel.

In the present reconstruction the proposition is a compact meeting point of three pieces of geometry already developed in Book I:

parallelism $\longrightarrow$ alternate-angle congruence $\longrightarrow$ SAS $\longrightarrow$ parallelism.

The first implication is Euclidean and uses the parallel axiom through the I.29 infrastructure. The final implication is the neutral converse alternate-angle theorem represented by I.27.

2 Classical Proof Architecture

In the classical configuration, let AB and CD be equal and parallel, with the joining segments AC and BD .

A diagonal is used to compare two triangles. Parallelism supplies a pair of congruent alternate angles, the given equality supplies one pair of corresponding sides, and the diagonal is common. SAS therefore gives the remaining side equality and the second required angle equality. The converse alternate-angle criterion then proves the second pair of lines parallel.

Schematically,

$AB \parallel CD \implies \text{alternate angles} \implies \text{SAS} \implies AC \cong BD \text{ and alternate angles} \implies AC \parallel BD.$
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The Lean proof preserves this architecture. What becomes explicit is the orientation information that a classical diagram carries silently.

3 Explicit Crossing Formulation

The lower-level theorem uses a crossing point M to encode the configuration:

$$A - M - D, \quad B - M - C.$$

Its documentation-level statement is

```
theorem euclid_proposition_33_crossing
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D M : Geo.Point)
  (hParallel : Geo.Parallel A B C D)
  (hCongruent : Geo.Congruent A B C D)
  (hAMD : Geo.Between A M D)
  (hBMC : Geo.Between B M C) :
  Geo.Parallel A C B D /\
  Geo.Congruent A C B D := by
  -- proof in Proposition33.lean
```

The two betweenness assumptions determine which sides of the transversal contain the four endpoints. From them the proof reconstructs the SameSide/OppositeSide and SameRay information required by the Hilbert angle interface.

4 Configuration Normalization

The main formal issue in I.33 is not SAS itself but orientation.

From

$$B - M - C$$

the proof obtains the incidence data for the transversal through B, M, C . Since $AB \parallel CD$, the endpoints A and D cannot lie on that transversal. Together with

$$A - M - D$$

this yields the required opposite-side relation.

An interior point of BC is then used to normalize the rays entering the angle statements. This makes it possible to apply the Euclidean alternate-angle theorem in exactly the orientation needed for SAS.

Thus the formal preparation is

$$\text{betweenness} \longrightarrow \text{incidence} \longrightarrow \text{side-of-line data} \longrightarrow \text{same-ray normalization}.$$

No additional Euclidean content is introduced in this stage.

5 The I.29 Step

The Euclidean direction of the parallel-angle theory is invoked through the established interface theorem for alternate angles of parallel lines. Conceptually it supplies

$$AB \parallel CD \implies \angle BAC \cong \angle DCB$$

after the required ray and side normalizations.

This is the only essentially Euclidean step in the proof. It is precisely where Hilbert Group IV enters.

6 The SAS Step

The two triangles determined by the diagonal are compared by SAS.

The proof uses:

- the given segment congruence $AB \cong CD$;
- the common diagonal;
- the angle congruence obtained from parallelism.

The resulting triangle-congruence object provides two pieces of information needed for the conclusion.

First,

$$AC \cong BD.$$

Second, it supplies a further angle congruence. After same-ray normalization, this is the alternate-angle statement required to prove

$$AC \parallel BD.$$

7 The I.27 Step

The second parallelism is obtained from the converse alternate-angle theorem.

This part is neutral: once the appropriate alternate angles are known congruent and the side configuration is normalized, no use of the Euclidean parallel axiom is required.

The architecture can therefore be written more sharply as

configuration normalization $\longrightarrow$ I.29 $\longrightarrow$ SAS $\longrightarrow$ I.27.
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8 Relationship with Earlier Propositions

Proposition I.33 combines the two directions of the parallel-angle theory developed immediately before it.

The forward direction

$$\text{parallel lines} \implies \text{congruent alternate angles}$$

belongs to the Euclidean layer and is represented by I.29.

The converse direction

$$\text{congruent alternate angles} \implies \text{parallel lines}$$

is represented by I.27 and belongs to the neutral layer.

I.33 is therefore a useful architectural checkpoint: a Euclidean theorem is used to enter a triangle congruence argument, and the conclusion is completed by a neutral converse theorem.

9 Hilbert and Book Zero Components Used

No new axiom is introduced for Proposition I.33.

The proof uses established infrastructure for:

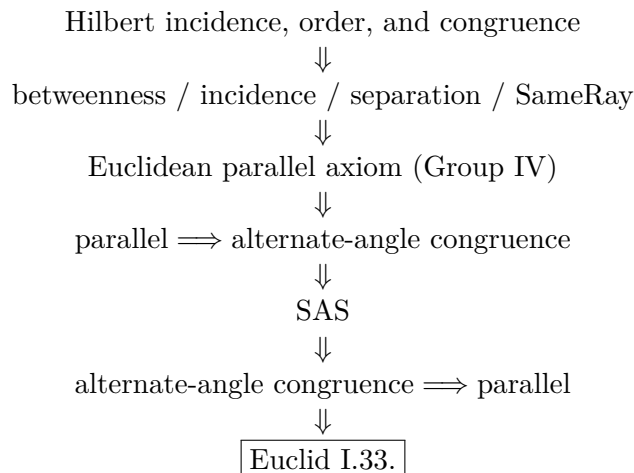
- incidence and line carriers;
- betweenness and its incidence consequences;
- separation by a line;
- SameSide and OppositeSide relations;
- SameRay transport;
- segment-congruence symmetries;
- non-collinearity;
- SAS triangle congruence;
- the Euclidean alternate-angle theorem;
- the converse alternate-angle theorem.

The API-level formulation also uses the existing parallelogram infrastructure, including the one-pair-parallel-and-congruent criterion and congruence of opposite sides of a parallelogram.

No I.33-specific helper is added to `HilbertInterface` or to Book Zero.

10 Dependency Summary

The formal dependency structure is



New axiom:	no
New proposition-specific helper:	no
Continuity:	no
Euclidean parallel axiom:	yes
Build status:	clean
sorry:	no

11 Conclusion

Euclid I.33 is mathematically short but formally revealing. Once the orientation hidden in the classical picture is exposed, the original proof mechanism survives intact:

$$\boxed{\text{I.29} \longrightarrow \text{SAS} \longrightarrow \text{I.27.}}$$

The proposition therefore confirms that the developed Hilbert interface can express not only individual classical constructions, but also the interaction between the Euclidean and neutral layers of Book I.